FATEC IPIRANGA

ANÁLISE E DESENVOLVIMENTO DE SISTEMAS

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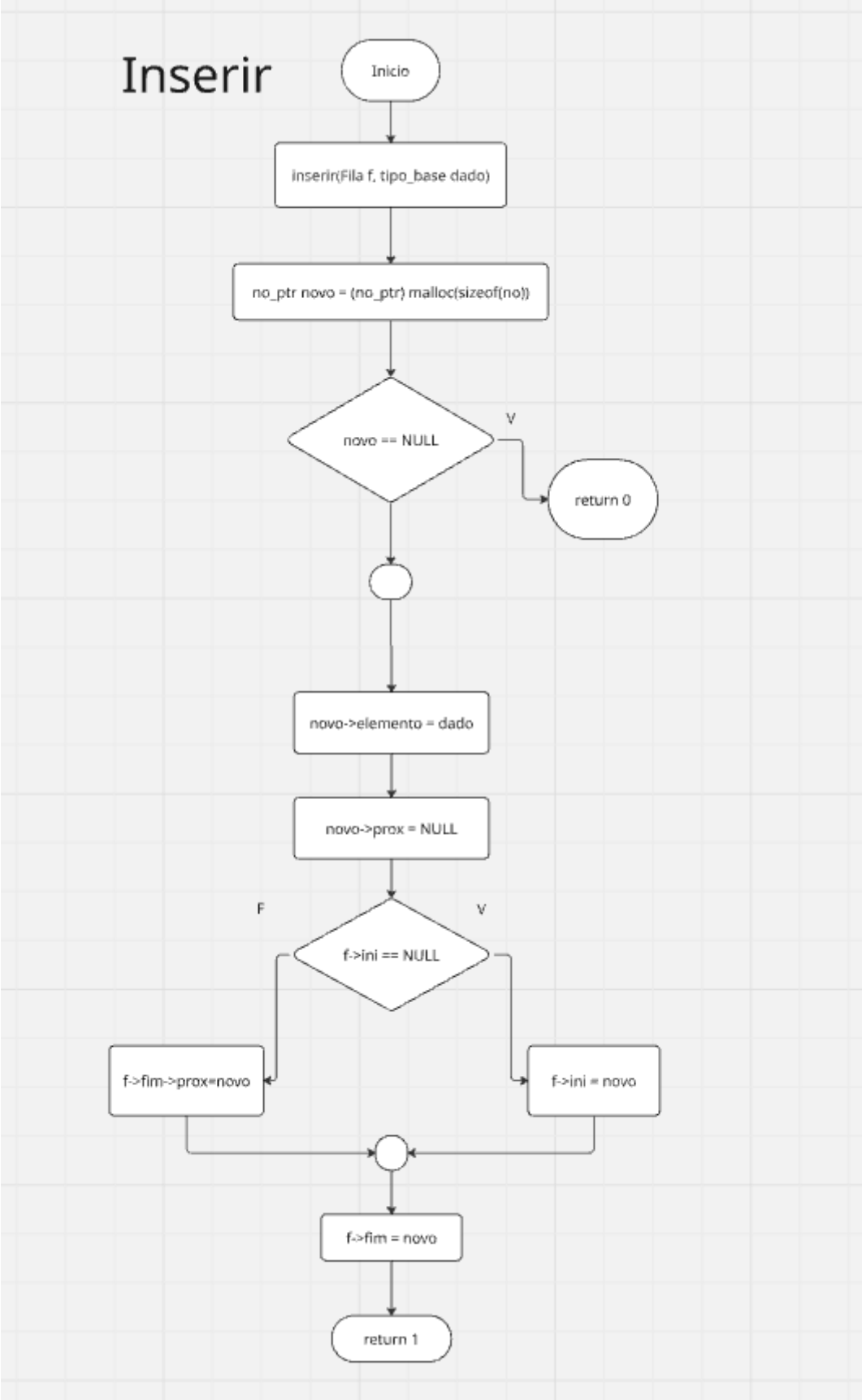
DIAGRAMA DE BLOCOS: ALGORÍTMO DE FILAS

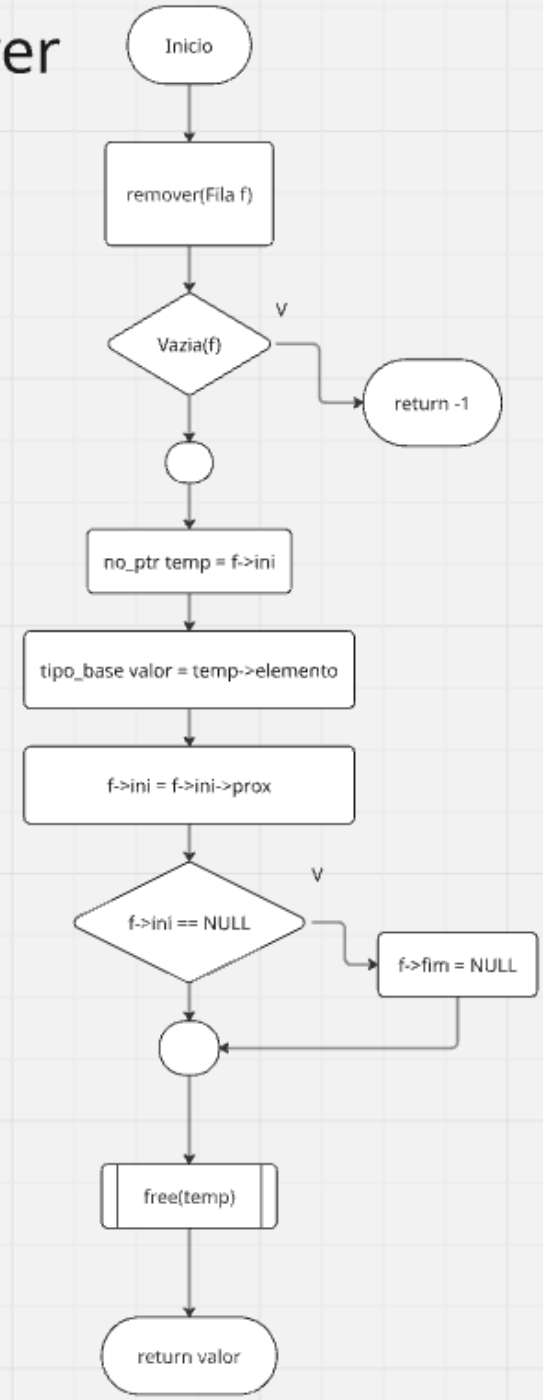
Estrutura de Dados

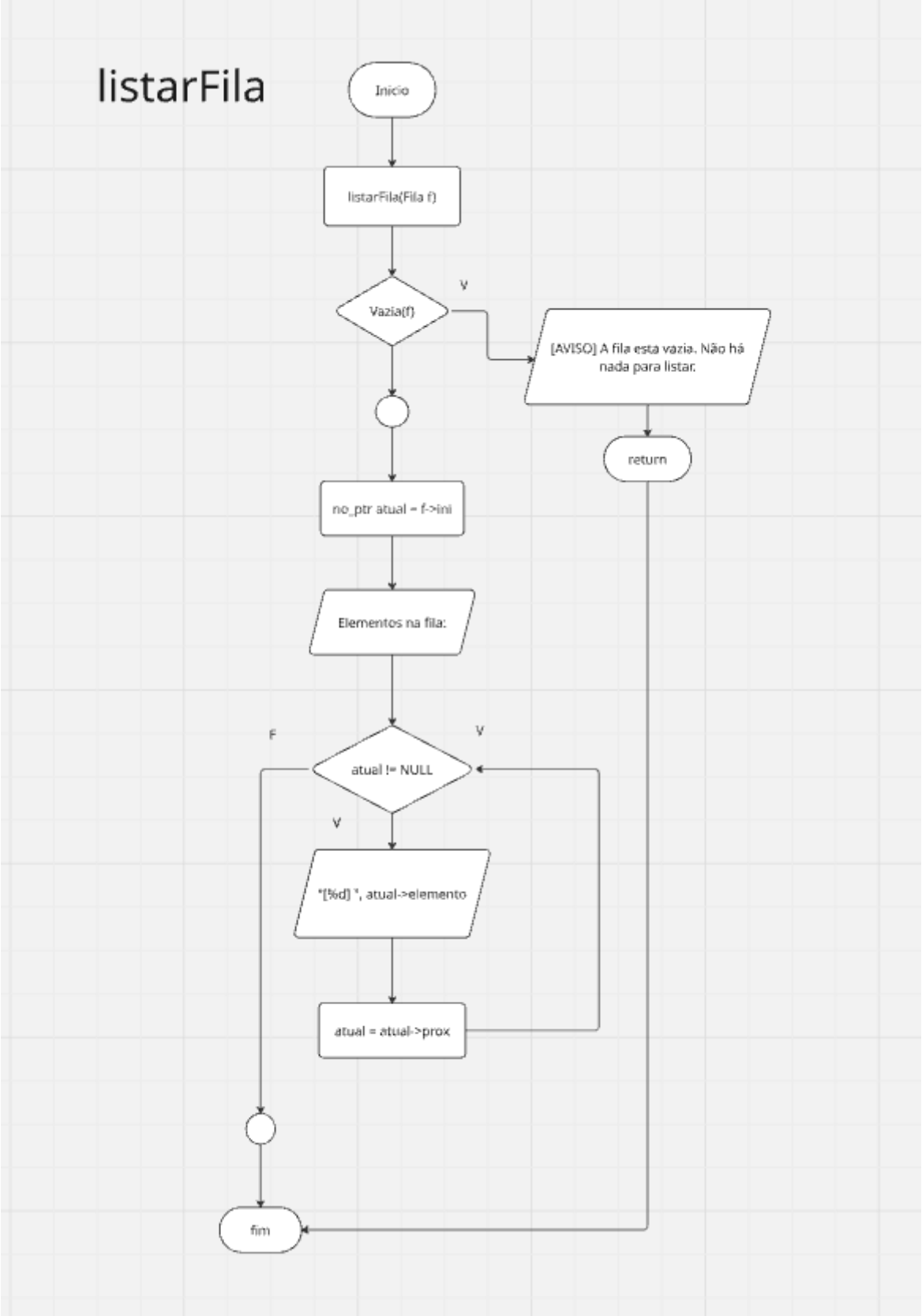
São Paulo

2026

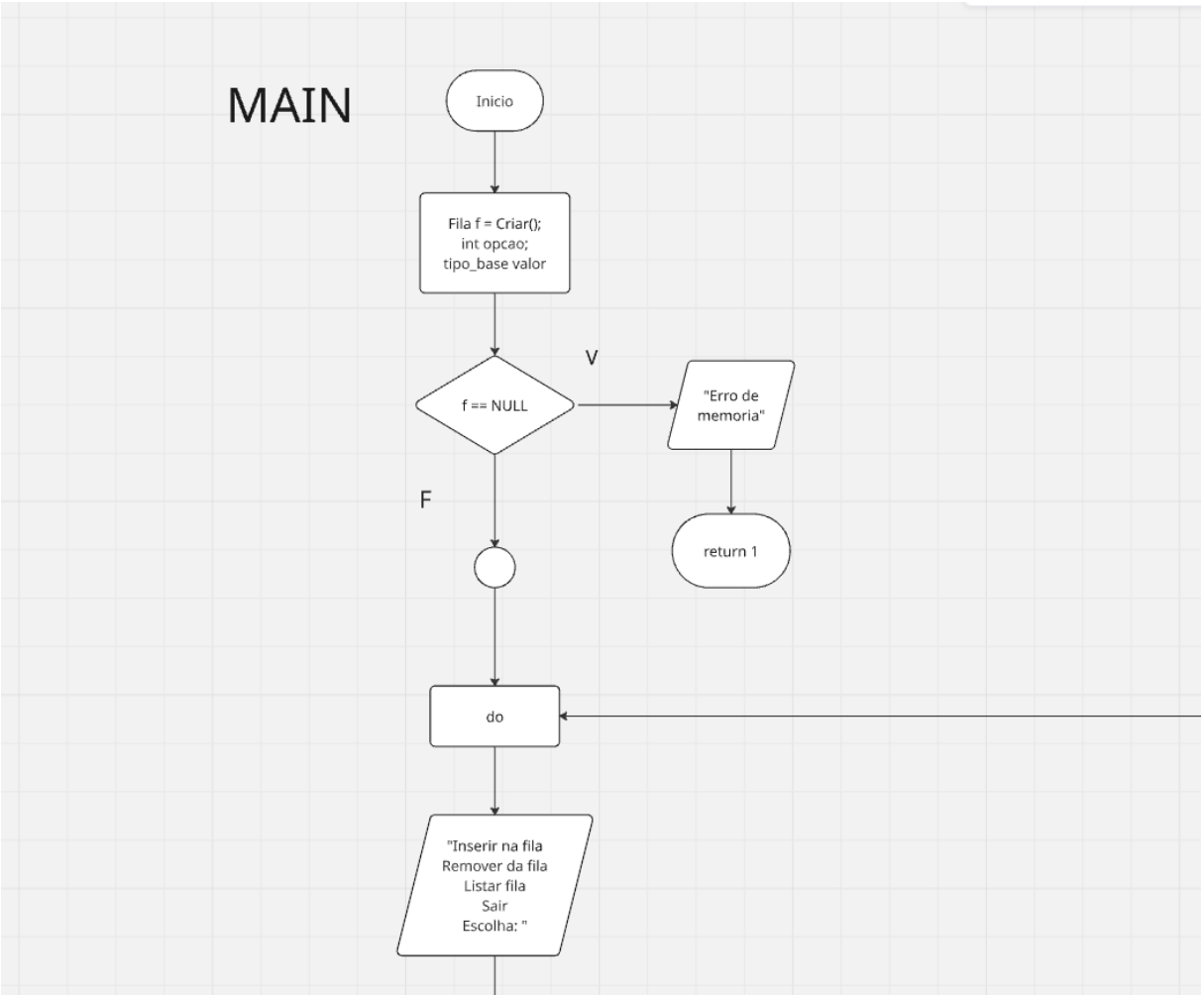

Inserir
Inicio
inserir(Fila f, tipo_base dado)
no_ptr novo = (no_ptr) malloc(sizeof(no))
novo == NULL
V
return 0
novo->elemento = dado
novo->prox = NULL
F
V
f->ini == NULL
f->fim->prox=novo
f->ini = novo
f->fim = novo
return 1

Remover

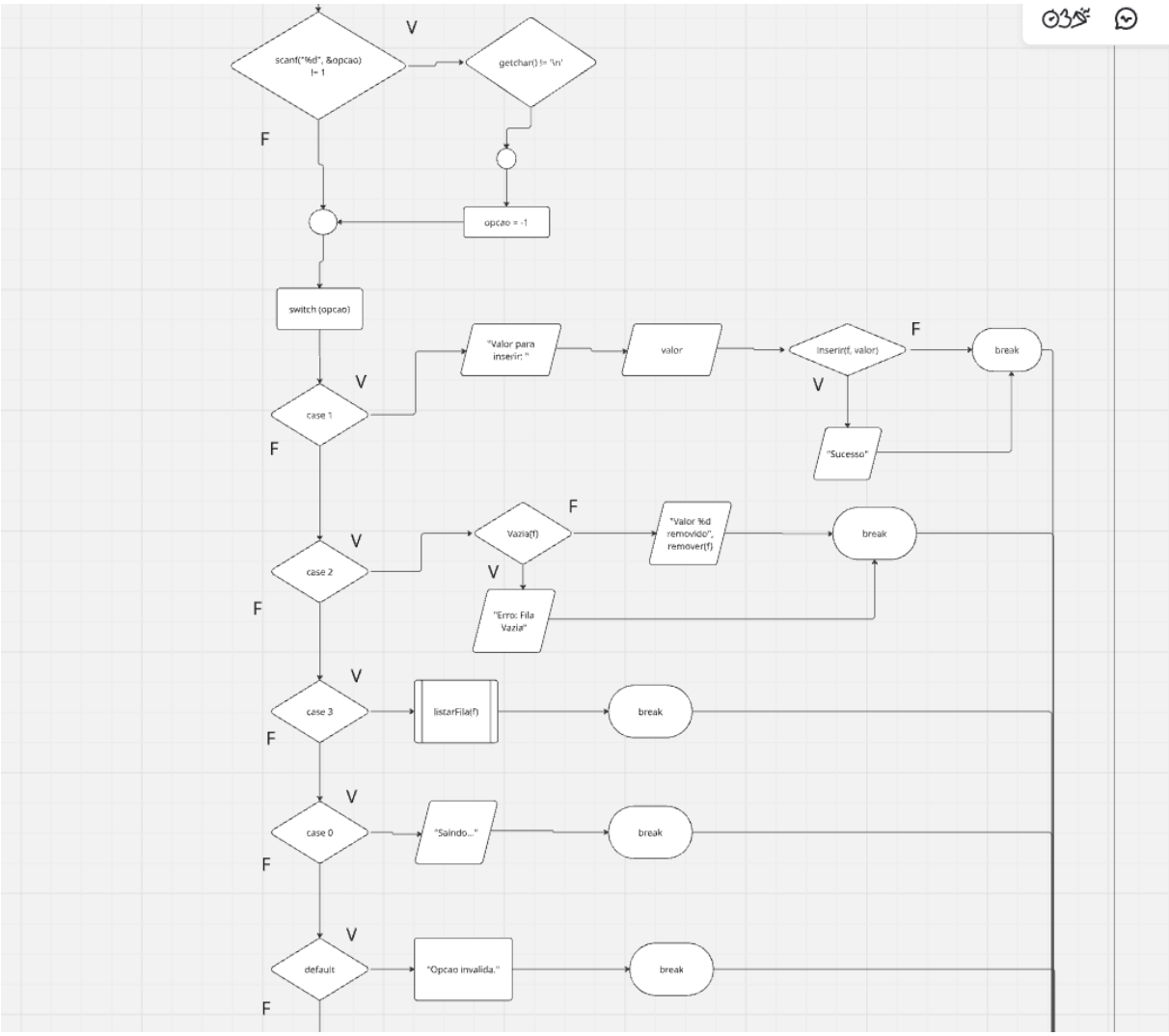

Inicio
remover(Fila f)
Vazia(f)
V
return -1
no_ptr temp = f->ini
tipo_base valor = temp->elemento
f->ini = f->ini->prox
f->ini == NULL
V
f->fim = NULL
free(temp)
return valor


listarFila
Inicio
listarFila(Fila f)
Vazia(f)
V
[AVISO] A fila esta vazia. Não há nada para listar.
return
no_ptr atual = f->ini
Elementos na fila:
F
V
atual != NULL
V
"[%d] ", atual->elemento
atual = atual->prox
fim

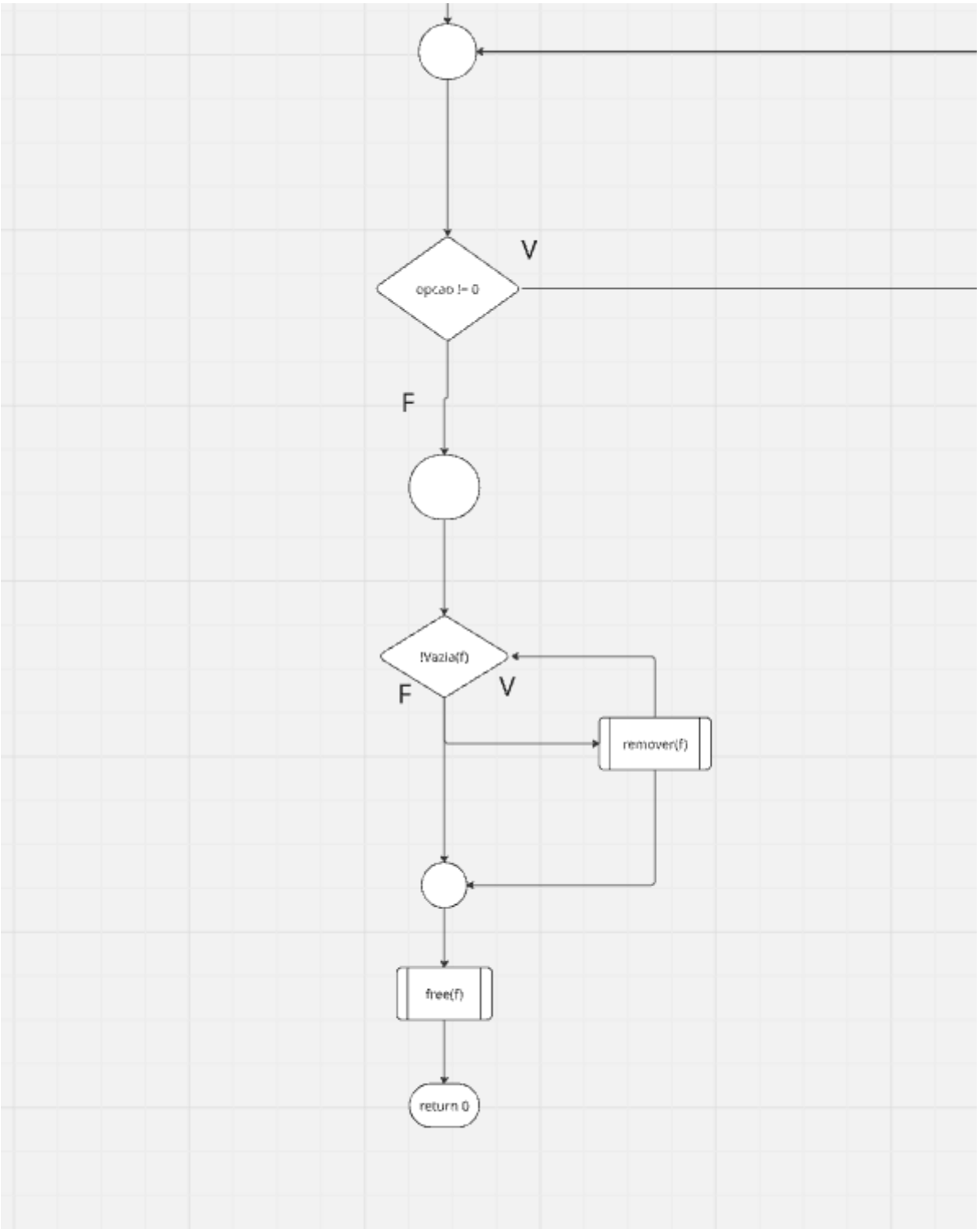
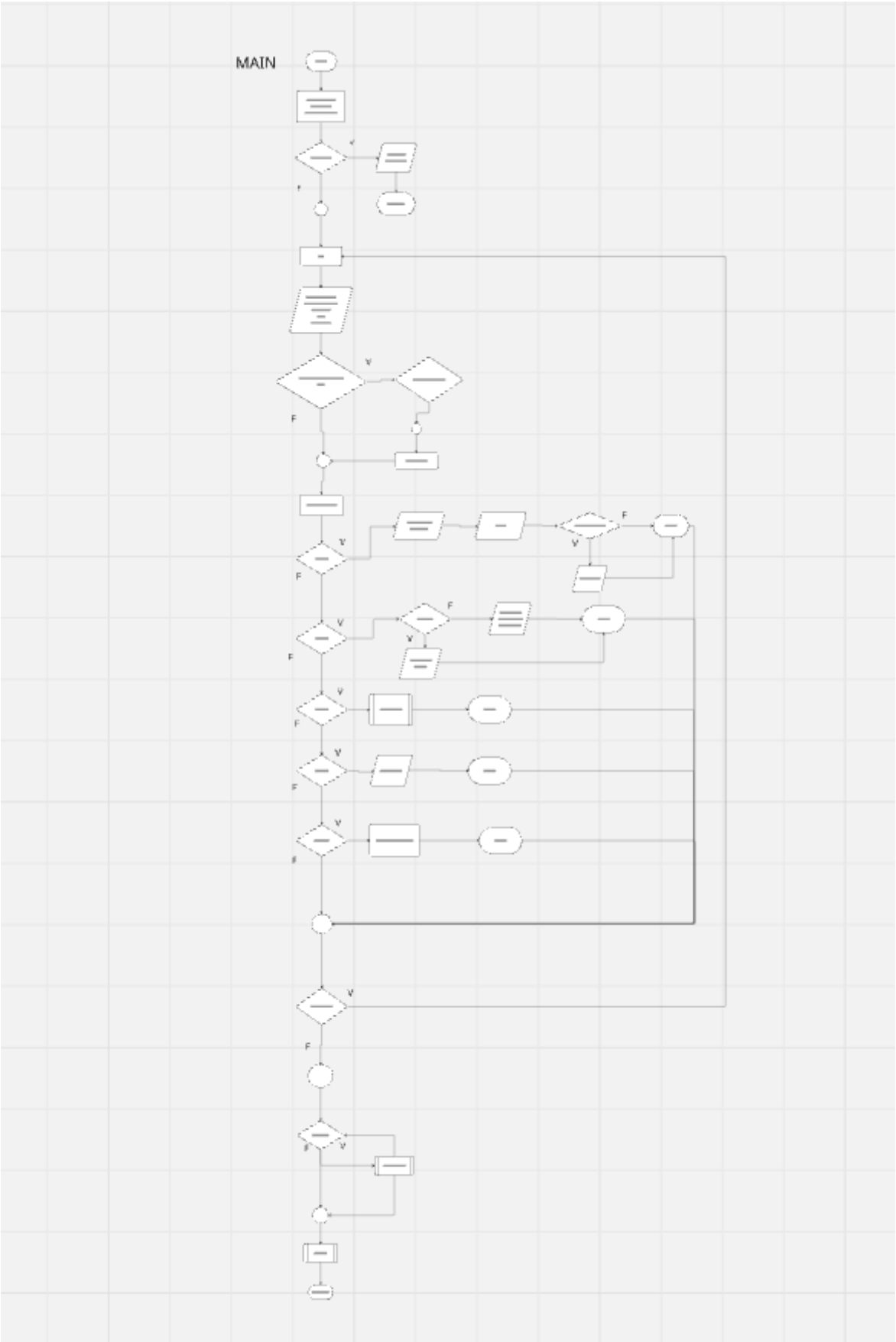
MAIN – PARTE 1


MAIN
Inicio
Fila f = Criar();
int opcao;
tipo_base valor
f == NULL
V
"Erro de memoria"
return 1
F
do
"Inserir na fila
Remover da fila
Listar fila
Sair
Escolha: "

MAIN - PARTE 2


scanf("%d", &opcao) != 1
V
getchar() != '\n'
F
opcao = -1
switch (opcao)
case 1
V
F
"Valor para inserir: "
valor
inserir(f, valor)
F
V
break
"Sucesso"
case 2
V
F
Vazia(f)
F
V
"Valor %d removido", remover(f)
break
"Erro: Fila Vazia"
case 3
V
F
listarFila(f)
break
case 0
V
F
"Saindo..."
break
default
V
F
"Opcao invalida."
break

MAIN – PARTE 3



ESTRUTURA DO MAIN